

Soham Khisa

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RESEARCH INTEREST

Machine Learning, Computer Vision; Natural Language Processing; Human-Computer Interaction

EDUCATION

University of Maryland, Baltimore County

Fall 2025 - Present

Doctor of Philosophy in Computer Science

Bangladesh University of Engineering and Technology (BUET)

2018–2023

Bachelor of Science in Computer Science and Engineering

Thesis: Knowledge Graph-Based Categorization of Newspaper Articles in a Newspaper Corpus [[Preprint](#)]

Advisor: Dr. Muhammad Masroor Ali

EXPERIENCE

1. Graduate Teaching Assistant

Aug 2025 – Present

Department of Computer Science and Electrical Engineering, UMBC

Course: CMSC421 - Principles of Operating Systems

Course: CMSC331 - Programming Languages

2. Research Assistant (Remote)

Sept 2023 – May 2025

Bowie State University, MD

Advisor: [Dr. Avijoy Chakma](#)

Research area: Machine learning in IoT

PUBLICATION AND PREPRINT

1. Avijoy Chakma, Adity Khisa, **Soham Khisa**, Jannatun Noor, Sharifa Sultana, “Re-educating Educated Ones: A Case Study on Chakma Language Revitalization in Chittagong Hill Tracts” **CHI ’26: Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems** [[Paper](#)]
2. Aunabil Chakma, Aditya Chakma, Masum Hasan, **Soham Khisa**, Chumui Tripura, and Rifat Shahriyar, “ChakmaNMT: Machine Translation for a Low-Resource and Endangered Language via Transliteration” [[Under review](#)] [[Paper](#)]
3. **Soham Khisa**, Avijoy Chakma, “VaCDA: Variational Contrastive Alignment-based Scalable Human Activity Recognition” **2025 International Conference on Machine Learning and Applications (ICMLA)** [[Paper](#)]

SERVICES

Emergency Reviewer, International Joint Conference on Artificial Intelligence (IJCAI) [[PDF](#)]

2025

PROJECTS

1. CNN From Scratch

Deep Learning Project

Libraries: numpy, opencv, matplotlib, tqdm, pandas, pickle, scipy

Architecture: Convolutional Neural Network (LeNet)

Developed a CNN-LeNet architecture from scratch without using any machine learning frameworks like Tensorflow or PyTorch. The model was trained and evaluated on the MNIST dataset.

[\[Code\]](#)

2. Retinal Disease Classification

Deep Learning Project

Frameworks & Libraries: numpy, matplotlib, pandas, PyTorch

Architecture: Restricted Boltzmann Machine, Convolutional Neural Network (AlexNet)

The project's objective is to use retinal images to recognize and classify various retinal illnesses, including age-related macular degeneration and diabetic retinopathy.

[\[Code\]](#)

3. Auc-dais

Software Development Project

Frameworks & DBMS: Spring Boot, React, Bootstrap, PostgreSQL

This is a web application for an online auction platform system that allows users to list items for auction, and bidders can browse and select the items they wish to bid on.

[\[Frontend-Code\]](#) [\[Backend-Code\]](#) [\[Video\]](#)

ACHIEVEMENTS AND AWARDS

Technical **Scholarship** by Bangladesh Government (2018-2023) for **outstanding results** in Higher Secondary School.

Chittagong Hill Tracts Development Board Scholarship in 2021

Rangamati Hill District Council Scholarship in 2020

Finalist - MIST Inter-university ICT Innovation Fest, 2021

Finalist - HackNSU season 3, 2021

SKILLS

Programming Languages: Python, Java, C, C++, C#, Assembly

Machine Learning Frameworks/Libraries: PyTorch, TensorFlow, NumPy, scikit-learn, Pandas, Keras

Database Management: SQL (Oracle, Postgres, MySQL, SQLite)

Software Development: Django, Spring Boot, React, JavaFX, Bootstrap

Version Control: Git, GitHub

Operating Systems: Linux, Windows, Mac

Scripting: L^AT_EX, HTML, Shell Script(Linux)

CERTIFICATION

Workshop on AI in Public Health at [CHRF](#), organized by [Dr. Yogesh Hooda](#) and [Dr. Senjuti Saha](#), supported by the [Gates Foundation](#).
[\[Certificate\]](#) June 1, 2024

Deep Learning Specialization by [DeepLearning.AI](#) on [Coursera](#)
[\[Certificate\]](#) Oct 30, 2020

Machine Learning Foundations: A Case Study Approach by the University of Washington on [Coursera](#) [\[Certificate\]](#) May 29, 2020

REFERENCE

[Dr. Avijoy Chakma](#) (Research Advisor)

[Dr. Muhammad Masroor Ali](#) (Undergrad Thesis Advisor)